

LAB EXPERIMENT RECORD

Set 7: Customer Demographics Analysis

Department	AIML
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyze customer demographics by visualizing the distribution of customer ages using a bar chart, the distribution of customers by gender using a pie chart, presenting the demographic information in a labelled table, and combining all three views in an interactive Tableau dashboard.

Dataset: Customer Demographics

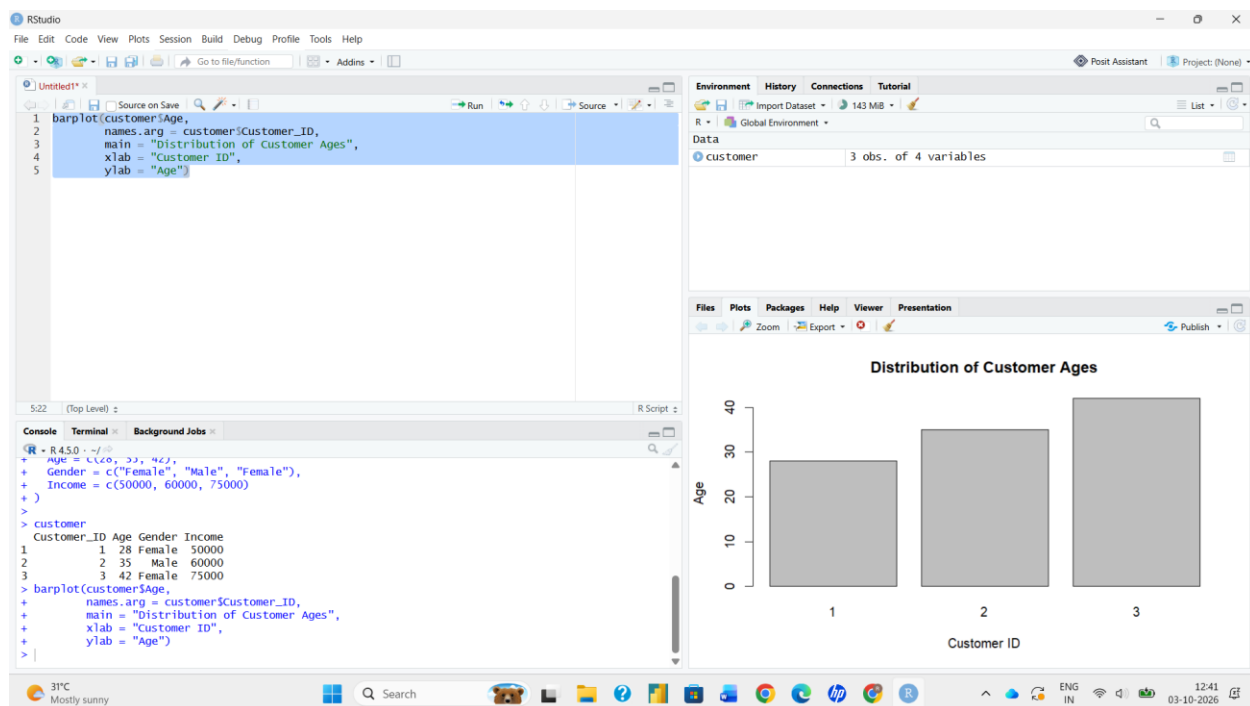
Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

PROGRAM 1: Bar Chart of Customer Ages

Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
# Dataset
customers <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

# Bar Chart
barplot(customers$Age,
  names.arg = customers$Customer_ID,
  xlab = "Customer ID",
  ylab = "Age (years)",
  main = "Distribution of Customer Ages",
  col = "skyblue")
```



Output 1: Customer Age Bar Chart

Observation: The ages of the three customers are 28, 35 and 42 years. Customer 3 is the oldest and Customer 1 is the youngest, and the average customer age is 35 years.

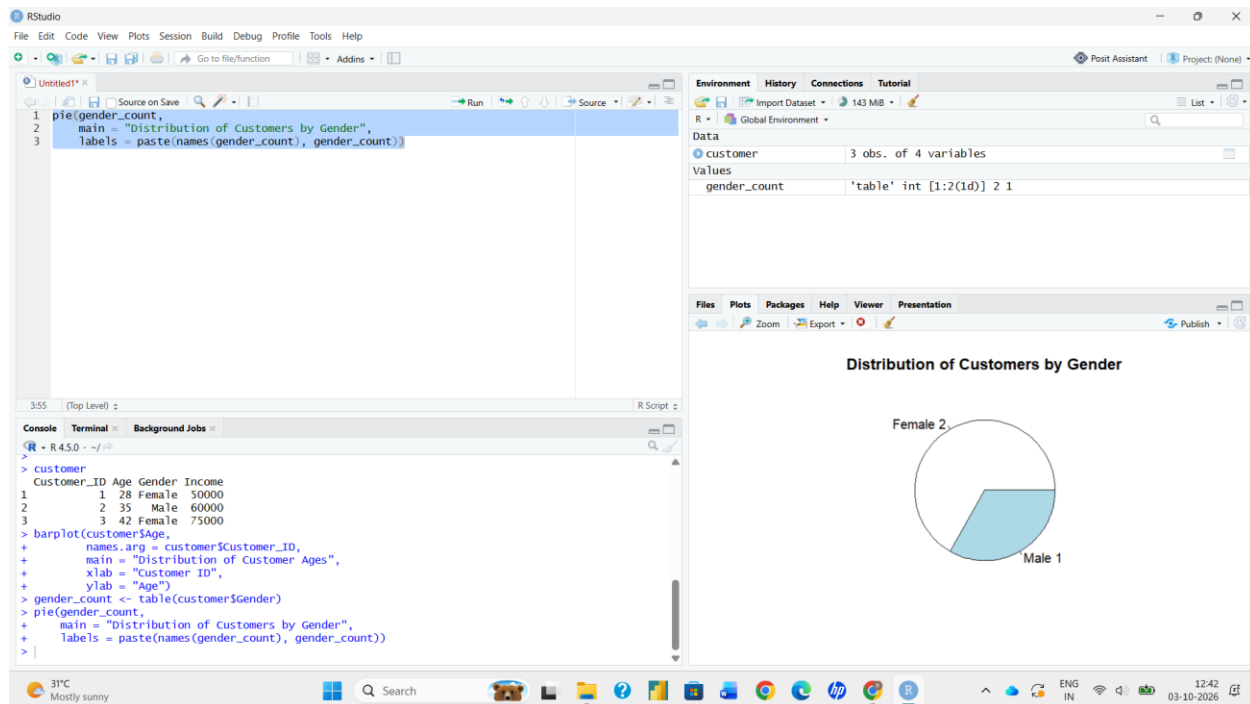
PROGRAM 2: Pie Chart of Customers by Gender

Generate a pie chart to represent the distribution of customers by gender.

```
# Count customers by gender
gender_count <- table(customers$Gender)

# Percentage labels
percent <- round(100 * gender_count / sum(gender_count), 1)
labels <- paste0(names(gender_count), " (", percent, "%)")

# Pie Chart
pie(gender_count,
    labels = labels,
    col = c("pink", "lightblue"),
    main = "Customer Distribution by Gender")
```



Output 2: Customer Gender Distribution Pie Chart

Observation: Two of the three customers are female (66.7%) and one is male (33.3%), so female customers form the larger share of this dataset.

PROGRAM 3: Table of Customer Demographic Information

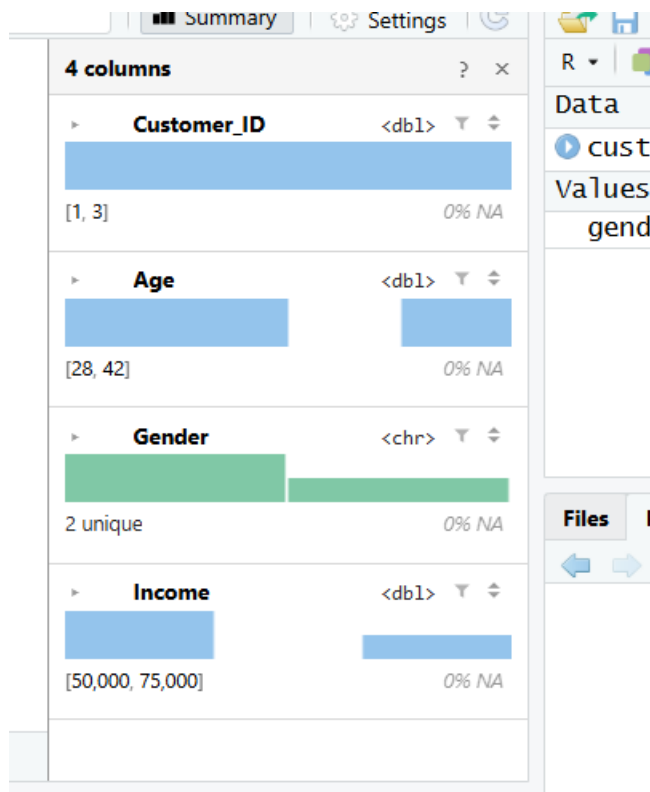
Build a table to show the demographic information for each customer. Label the table elements.

```
# Build and label the table
demographics <- customers
colnames(demographics) <- c("Customer ID", "Age", "Gender", "Income (in $)")

print(demographics)
```

```
Customer ID Age Gender Income (in $)
1          1  28 Female    50000
2          2  35  Male    60000
3          3  42 Female    75000
```

Output 3(a): R Console Output



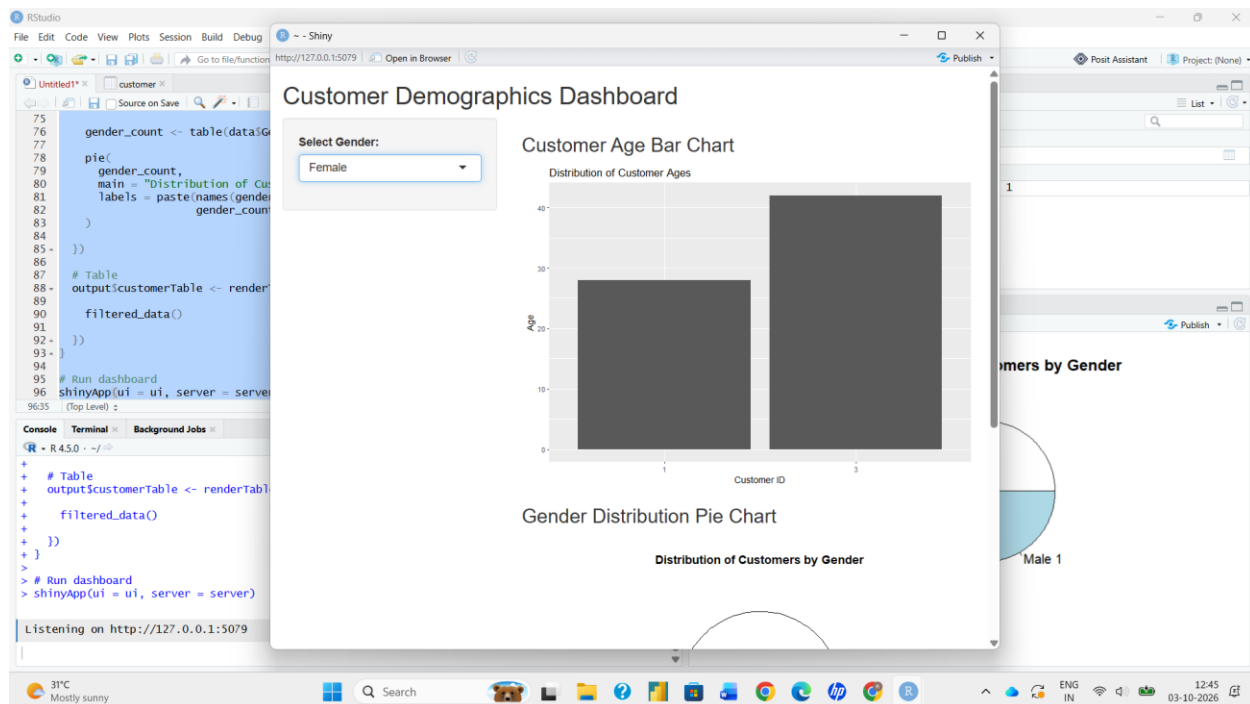
Output 3(b): Customer Demographic Information Table

Observation: The table lists the Customer ID, Age, Gender and Income of every customer. Income increases with age, from \$50,000 for the 28-year-old customer to \$75,000 for the 42-year-old customer, with an average income of about \$61,667.

PROGRAM 4: Interactive Tableau Dashboard

Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

1. Open Tableau and choose Connect > Text file (or Microsoft Excel), then select the customer demographics file (Customer ID, Age, Gender, Income).
2. Open Sheet 1 and create the bar chart: drag Customer ID to Columns and Age to Rows (set Age to SUM or to a Dimension). Add labels, rename the sheet to “Customer Ages” and title it.
3. Open Sheet 2 and create the pie chart: change the Marks type to Pie, drag Gender to Color and Customer ID (Count) to Angle. Show percentage labels and rename the sheet to “Gender Distribution”.
4. Open Sheet 3 and create the table: drag Customer ID, Age, Gender and Income to Rows (or double-click each field) to display the text table. Rename the sheet to “Customer Details”.
5. Click New Dashboard. Drag the three sheets onto the dashboard canvas and arrange them as shown in the output.
6. Right-click the Gender field > Show Filter, then use the filter’s dropdown > Apply to Worksheets > All Using This Data Source.
7. Select the pie chart and click “Use as Filter” so that clicking a gender slice highlights the matching bars and table rows.
8. Add a dashboard title, and test the interactivity by selecting Female or Male.



Output 4: Customer Demographics Tableau Dashboard

Observation: The dashboard shows ages, gender share and individual records together. Selecting a gender filters all three views at once, which lets the user explore the demographics interactively.

Procedure

1. Create the customer data frame with Customer_ID, Age, Gender and Income in R.
2. Use `barplot()` to plot the customer ages with labelled axes and a title.
3. Use `table()` to count customers by gender and `pie()` to draw the gender distribution.
4. Rename the data frame columns with `colnames()` and print the labelled demographic table.
5. Load the data into Tableau, build the bar chart, pie chart and table on separate worksheets.
6. Combine the worksheets on a dashboard and add the Gender filter and the Use as Filter action.

Result

The customer demographics were successfully visualized using a bar chart of ages, a pie chart of gender distribution and a labelled demographic table in R. An interactive Tableau dashboard combining all three views was created for exploring the customer data.

Key Functions / Tools Used

Function / Tool	Purpose
<code>data.frame()</code>	Creates the customer dataset.
<code>barplot()</code>	Creates the bar chart of customer ages.
<code>table()</code>	Counts customers in each gender category.
<code>pie()</code>	Creates the gender distribution pie chart.
<code>colnames()</code> / <code>print()</code>	Labels and displays the demographic table.
Tableau Dashboard	Combines the worksheets into one interactive view.
Tableau Filters	Allow the user to explore the data by gender.